SKILLBRIDGE PLATFORM

System Installation & Deployment Manual

A Comprehensive Administrator and Developer Guide for Installing, Configuring, Orchestrating, and Testing the SkillBridge Multi-Tier Ecosystem on Windows

Target Environment:	Microsoft Windows 10 / Windows 11 (64-bit)
Platform Version:	SkillBridge v1.0.0 (Release Edition)
Node.js Runtime:	v18.x – v24.x LTS (Tested on v24.0.0)
Python Runtime:	Python 3.10 – 3.14 (Virtual Environment)
Database Engine:	MongoDB Community Server 7.x (Port 27017)
Target Audience:	System Administrators, Evaluators, DevOps, Developers
Document Release:	November 2026

DEPLOYMENT DIRECTIVE

This installation guide covers the exact operational setup of the SkillBridge repository located at **C:\Skill Bridge**. Follow every prerequisite check and sequential step to ensure fault-tolerant cross-service intercommunication between the React client, Express API Gateway, FastAPI AI engine, and MongoDB daemon.

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1. Architecture Overview & System Topology

SkillBridge is structured as a decoupled, multi-tier enterprise application designed for high performance, modularity, and strict separation of concerns. The platform segregates user presentation, business logic orchestration, intensive AI text extraction, and ACID-compliant document persistence into four dedicated runtime tiers.

Tier	Component / Service	Technology Stack	Port	Core Responsibility
Tier 1	Presentation Layer (SPA Client)	React 18.3, TypeScript, Vite 6.1, Tailwind CSS	5173	User interfaces, role guards, reactive state management, file uploads.
Tier 2	API Gateway & App Server	Node.js 24, Express 4.21, Mongoose, JWT, Multer	5000	Authentication, RBAC enforcement, file storage, database orchestration.
Tier 3	AI Microservice Engine	Python 3.12, FastAPI 0.110, Uvicorn, WinOCR, pdfplumber	8000	Text extraction, OCR, skill parsing, normalization, credibility scoring.
Tier 4	Persistence Layer	MongoDB Community Server 7.x	27017	JSON document persistence for users, profiles, evidence, and audit logs.

Inter-Service Communication Flow

Communication between tiers adheres to strict directional flow patterns:

- **Client-to-Gateway (HTTP/JSON):** The React SPA communicates solely with the Express API Gateway on port 5000 via Axios over REST endpoints prefixed with /api or /api/v1. Bearer JWT tokens accompany all authenticated requests.
- **Gateway-to-AI Engine (Internal REST):** The Express backend proxies document extraction requests to FastAPI on port 8000 (http://127.0.0.1:8000/api/ai/extract-text). The client never directly calls the Python microservice.
- **Gateway-to-Database (Mongoose ODM):** Express maintains a persistent connection pool to MongoDB on port 27017 via mongodb://127.0.0.1:27017/skillbridge.

SECURITY ARCHITECTURE DIRECTIVE

The architecture enforces strong zero-trust boundaries: the AI service operates as an internal backend worker, and frontend clients cannot access the database or AI endpoints directly without passing through Express authentication.

2. Prerequisites & System Requirements

Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 (Dual-Core)	Intel Core i5 / AMD Ryzen 5 (Quad-Core or higher)
System Memory (RAM)	8 GB RAM	16 GB RAM (optimal for parallel OCR and node runtimes)
Storage Space	5 GB free disk space	10 GB+ SSD free space (for PDF storage and models)
Display Resolution	1366 x 768	1920 x 1080 (Full HD) or higher

Operating System & Software Runtime Matrix

Software Package	Supported Versions	Verification Command	Purpose
Operating System	Windows 10 / Windows 11 (64-bit)	winver	Host OS with PowerShell 5.1+
Node.js & npm	Node v18.0.0 – v24.x LTS	node -v ; npm -v	Executes Express backend & Vite build tool
Python	Python 3.10.x – 3.14.x (64-bit)	python --version	FastAPI microservice, WinOCR, Pillow, pdfplumber
MongoDB Server	Community Edition v6.0 – v7.0+	mongod --version	Document database engine running on port 27017
Git for Windows	Git 2.40+	git --version	Version control and repository management

Pre-Flight System Check Command

Before beginning the installation, open PowerShell as Administrator and run the following pre-flight diagnostics:

PowerShell Pre-Flight Diagnostics

```
# Execute Pre-Flight Software Verification in PowerShell
Write-Host 'Checking Node.js:' -NoNewline; node -v
Write-Host 'Checking npm:'     -NoNewline; npm -v
Write-Host 'Checking Python:'  -NoNewline; python --version
Write-Host 'Checking MongoDB:' -NoNewline; mongod --version | Select-String 'db version'
Write-Host 'Checking Git:'     -NoNewline; git --version
```

3. Step-by-Step Clean Installation

Follow these steps sequentially to configure the complete SkillBridge environment from a clean clone of the repository.

Step 3.1: Repository Structure & Workspace Layout

The project workspace is rooted at C:\Skill Bridge. Confirm that your directory structure contains the following core subdirectories:

SkillBridge Root File Layout

```
C:\Skill Bridge\
■■■ frontend/                # React 18 + Vite SPA client
■■■ backend/                 # Express REST API gateway
■■■ ai-service/              # FastAPI microservice & virtualenv
■■■ scripts/                 # Orchestration, shutdown, and audit scripts
■■■ documentation/           # High-resolution screenshots and assets
■■■ start_skillbridge.bat    # Windows one-click desktop launcher
■■■ stop_skillbridge.bat     # Windows graceful shutdown launcher
■■■ User Manual.pdf          # User Manual deliverable
■■■ Installation Manual.pdf  # Installation Manual deliverable
```

Step 3.2: MongoDB Database Installation & Service Startup

SkillBridge requires MongoDB Community Edition to be installed as a Windows Service or running locally on port 27017.

1. Download MongoDB Community Server from https://www.mongodb.com/try/download/community.
2. Run the installer and ensure the option **'Install MongoDB as a Service'** is checked.
3. Verify the service is active or start it manually via PowerShell:

PowerShell MongoDB Service Control

```
# Check MongoDB service status and start if stopped
Get-Service -Name MongoDB
if ((Get-Service -Name MongoDB).Status -ne 'Running') {
    Start-Service -Name MongoDB
}
# Verify port 27017 is actively listening
Get-NetTCPConnection -LocalPort 27017 -State Listen
```

Step 3.3: AI Microservice Python Virtual Environment Setup

The AI microservice handles text extraction and certification analysis. It must be initialized with its own isolated Python virtual environment:

1. Open PowerShell and navigate to the ai-service directory.
2. Create a virtual environment named venv.
3. Upgrade pip and install dependencies from requirements.txt.
4. Install reportlab for PDF documentation generation.

PowerShell Python Environment Setup

```
cd "C:\Skill Bridge\ai-service"
python -m venv venv
& .\venv\Scripts\python.exe -m pip install --upgrade pip
& .\venv\Scripts\python.exe -m pip install -r requirements.txt
& .\venv\Scripts\python.exe -m pip install reportlab
```

Step 3.4: Express API Gateway Setup & Database Seeding

The Express backend manages authentication, data models, and API endpoints. Follow these steps to configure dependencies and populate initial roles:

1. Navigate to backend and install npm dependencies.
2. Verify that backend\.env exists with the proper configuration parameters.
3. Run the database seed script to populate default users, roles, skills, and sample entities.

Backend Dependency Installation & Database Seeding

```
cd "C:\Skill Bridge\backend"
npm install
node src/utils/seed.js
```

Configuration File Reference: backend\.env

backend/.env Production Configuration

```
PORT=5000
NODE_ENV=development
MONGODB_URI=mongodb://127.0.0.1:27017/skillbridge
JWT_SECRET=skillbridge_local_dev_secret_key_2026_mit_pw
JWT_EXPIRES_IN=7d
AI_SERVICE_URL=http://127.0.0.1:8000
AI_SERVICE_TIMEOUT_MS=8000
FRONTEND_URL=http://localhost:5173
UPLOAD_DIR=uploads
MAX_FILE_SIZE_MB=5
```

Step 3.5: React Frontend Client Setup & Build Verification

The Vite single-page application provides the unified interface. Install its dependencies and verify the TypeScript build:

1. Navigate to frontend and install all npm packages.
2. Verify that frontend\.env contains VITE_API_BASE_URL=http://localhost:5000/api.
3. Execute a TypeScript compile and Vite production build test to confirm zero syntax or bundling errors.

Frontend Dependency Installation & Build Test

```
cd "C:\Skill Bridge\frontend"
npm install
npm run build
```

VERIFICATION NOTE

The build command executes tsc -b && vite build. If this completes without errors, the frontend code is 100% type-safe and ready for local development or production hosting.

4. Application Orchestration & Service Management

SkillBridge features automated multi-process orchestration. You can launch and manage all three services (FastAPI, Express Gateway, React Frontend) alongside MongoDB using single-click launchers or dedicated terminal sessions.

Method A: One-Click Desktop Shortcut Launcher

Two desktop shortcuts are available on the system desktop for effortless zero-terminal launching and stopping:

- **SkillBridge – Start Application:** Double-clicking this shortcut triggers start_skillbridge.bat, which initiates pre-flight checks, verifies MongoDB, launches all three services in labelled windows, tracks their PIDs, waits for readiness, and automatically opens the browser at http://localhost:5173.
- **SkillBridge – Stop Application:** Double-clicking this shortcut executes stop_skillbridge.bat, which safely closes all child processes without terminating unrelated system programs.

Method B: Automated Batch & PowerShell Orchestration Script

To execute the orchestration from the terminal, run start_skillbridge.bat from the project root. The script performs the following 6 sequential stages:

Stage	Operation	Target & Logic
Stage 1/6	Environment Inspection	Verifies Node.js, npm, Python venv, and node_modules exist.
Stage 2/6	MongoDB Service Verification	Tests port 27017. If inactive, attempts Start-Service -Name MongoDB.
Stage 3/6	Port Conflict Check	Checks ports 8000, 5000, 5173 to warn or handle active bindings.
Stage 4/6	Service Spawning & PID Tracking	Spawns separate PowerShell windows for AI, Backend, and Frontend. Writes PIDs to .skillbridge_pids.json.
Stage 5/6	Health Polling & Readiness Check	Polls http://127.0.0.1:8000/health, http://localhost:5000/api/health, and http://localhost:5173.
Stage 6/6	Browser Auto-Launch	Launches default Windows browser targeting http://localhost:5173.

Method C: Graceful Application Shutdown Utility

To terminate the running instance safely, run stop_skillbridge.bat or execute:

PowerShell Controlled Graceful Shutdown

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File ".\scripts\stop_skillbridge.ps1"
```

The stopper reads .skillbridge_pids.json and executes taskkill /F /T /PID <id> to terminate the exact child process trees (FastAPI, Express, Vite) without killing MongoDB or unrelated terminal sessions.

Method D: Manual Multi-Terminal Development Launch

If you are actively developing or debugging code, you may prefer launching each service in its own dedicated terminal session to view live stack traces and logs:

Terminal 1 — MongoDB Database Daemon

Terminal 1: MongoDB Daemon

```
# Terminal 1: Ensure MongoDB service is running
net start MongoDB
```

Terminal 2 — AI Microservice Engine (FastAPI)

Terminal 2: FastAPI Microservice (Port 8000)

```
# Terminal 2: FastAPI Microservice
cd "C:\Skill Bridge\ai-service"
& .\venv\Scripts\python.exe -m uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload
```

Terminal 3 — API Gateway & App Server (Express)

Terminal 3: Express Backend Gateway (Port 5000)

```
# Terminal 3: Express REST API Server
cd "C:\Skill Bridge\backend"
node src/server.js
```

Terminal 4 — Presentation Client (React Vite)

Terminal 4: React Vite Dev Server (Port 5173)

```
# Terminal 4: React Vite Single Page App
cd "C:\Skill Bridge\frontend"
npm run dev
```

SEQUENCE ADVISORY

When running in multi-terminal mode, ensure you launch in the exact sequence: MongoDB -> AI Engine -> Express Gateway -> React Vite. This ensures Express can resolve database and AI microservice health checks immediately upon boot.

5. Default User Accounts & Access Personas

SkillBridge comes pre-seeded with dedicated demonstration accounts representing each distinct operational role. Use these credentials for evaluation, functional verification, and administrative configuration.

Role Persona	Email Address	Password	Primary Route	Operational Scope
Student	student@skillbridge.edu	Student@123	/dashboard	Evidence portfolio, AI analysis, skill gaps, readiness score.
Faculty Guide	faculty@skillbridge.edu	Faculty@123	/faculty	Assigned students, review queue, log book, endorsements.
Administrator	admin@skillbridge.edu	Admin@123	/admin	Program oversight, user audits, system settings, analytics.
Recruiter	recruiter@skillbridge.edu	Recruiter@123	/recruiter	Candidate search, talent pipeline, verified skill filters.
Real Student (Verified)	omingale26@gmail.com	Student@123	/dashboard	12 verified credentials, 25 extracted skills, 80/100 readiness.

Role-Based Access Control (RBAC) Architecture

Access control is enforced at both the client and server levels:

- **Frontend Route Guards:** RoleRoute and ProtectedRoute components inspect the decoded JWT token upon navigation. Users attempting to access unauthorized routes are redirected immediately to /access-denied.
- **Backend Middleware:** authorize('student'), authorize('faculty'), and authorize('admin') middleware intercept every API request. Unauthorized requests receive an explicit 403 Forbidden HTTP response.

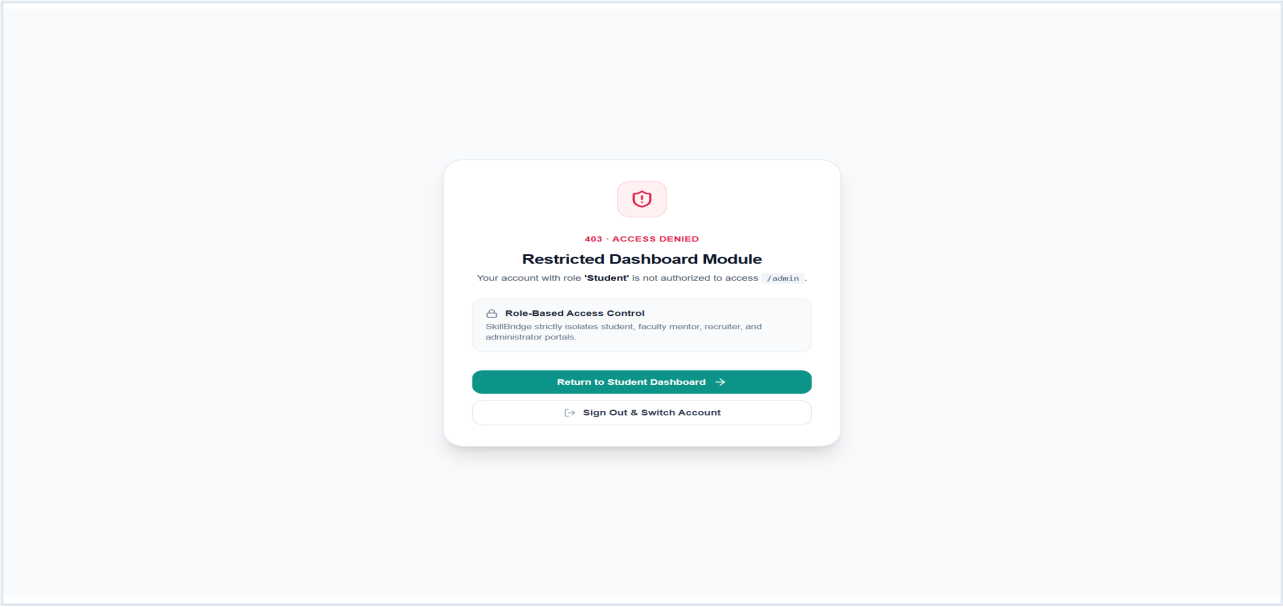


Figure: Role-Based Security Guard: Access Denied (403 Forbidden) Screen with Back-to-Safety Navigation

6. Comprehensive Verification & Diagnostic Testing

SkillBridge includes automated diagnostic test scripts to verify authentication, role segregation, and API authorization boundaries immediately following installation.

Test Suite 1: Authentication & JWT Verification

To verify that all seeded accounts authenticate cleanly and receive valid JWT tokens, run:

Authentication Test Execution

```
node scripts/test_auth_logins.js
```

Expected Outcome: Returns [PASS] for all 4 roles (Student, Faculty, Admin, Recruiter), verifying HTTP 200 responses, token issuance, and correct user role claims.

Test Suite 2: Role-Based Routing & Dashboard Matrix

To verify that each role navigates to its dedicated dashboard without leaking views across personas, run:

Role Routing Test Execution

```
node scripts/test_role_based_system.js
```

Expected Outcome: Validates that Students route to /dashboard, Faculty to /faculty, Admins to /admin, and Recruiters to /recruiter.

Test Suite 3: API Endpoint Access Control & 403 Guards

To verify strict backend authorization enforcement across protected endpoints, run:

RBAC Access Control Test Execution

```
node scripts/test_api_access_control.js
```

Expected Outcome: Confirms that cross-role access (e.g. a Student calling Admin endpoints or Faculty accessing student-only mutators) returns HTTP 403 Forbidden.

Direct HTTP Health Check Verification

You can verify service readiness directly from PowerShell using Invoke-RestMethod:

PowerShell Health Check Probes

```
# Check FastAPI AI Microservice Health
(Invoke-RestMethod -Uri 'http://127.0.0.1:8000/health').status
# Expected: 'ok'

# Check Express Backend Gateway Health
(Invoke-RestMethod -Uri 'http://localhost:5000/api/health').status
# Expected: 'ok' or HTTP 200
```

7. Troubleshooting, Fault Resolution & FAQs

This section details solutions to common environmental and runtime issues encountered during installation or execution.

Issue 1: Port Conflict (Port 5000, 5173, or 8000 Already in Use)

Symptom: Express fails with EADDRINUSE: address already in use :::5000, or FastAPI fails to bind to port 8000.
Resolution: Identify the process occupying the port and terminate it:

PowerShell Port Conflict Resolution

```
# Identify PID on port 5000
Get-NetTCPConnection -LocalPort 5000 -State Listen | Select-Object OwningProcess
# Force terminate process by PID
Stop-Process -Id  -Force
```

Issue 2: MongoDB Connection Refused (ECONNREFUSED 127.0.0.1:27017)

Symptom: Backend logs show MongoServerSelectionError: connect ECONNREFUSED 127.0.0.1:27017.
Resolution: Check whether the MongoDB Windows service is running. If not, start it manually:

MongoDB Service Startup

```
Start-Service -Name MongoDB
# If MongoDB is not registered as a service, start daemon manually:
mongod --dbpath="C:\data\db"
```

Issue 3: PowerShell Script Execution Policy Restriction

Symptom: Running start_skillbridge.ps1 fails with File cannot be loaded because running scripts is disabled on this system.
Resolution: Set the PowerShell execution policy for the current user or bypass it via the launcher batch file:

PowerShell Execution Policy Adjustment

```
# Option 1: Always launch via start_skillbridge.bat (uses -ExecutionPolicy Bypass)
# Option 2: Set execution policy in PowerShell once:
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Issue 4: Python Virtual Environment Missing or Corrupt

Symptom: Launcher reports [ERROR] Python virtual environment not found at C:\Skill Bridge\ai-service\venv.
Resolution: Recreate the virtual environment and reinstall packages:

Python Virtualenv Reinitialization

```
cd "C:\Skill Bridge\ai-service"
python -m venv venv
& .\venv\Scripts\python.exe -m pip install -r requirements.txt reportlab
```

8. Database Backup, Restoration & Maintenance

To safeguard student profiles, uploaded certifications, extracted skills, and audit trails, execute regular database dumps and preserve the local file upload repository.

Database Dump Procedure

Use MongoDB Database Tools (mongodump) to create an archive of the skillbridge database:

Database Backup Command

```
# Export SkillBridge database to a timestamped backup folder
$backupDate = Get-Date -Format 'yyyyMMdd_HHmmss'
mongodump --db skillbridge --out "C:\Skill Bridge\backups\mongo_$backupDate"
```

Database Restoration Procedure

To restore a previously archived state or transfer data to a new evaluation workstation:

Database Restore Command

```
# Restore SkillBridge database from backup archive
mongorestore --db skillbridge --drop "C:\Skill Bridge\backups\mongo_20261105_120000\skillbridge"
```

Preserving Uploaded Documents & Certificates

In addition to MongoDB collections, student documents (PDFs, images) are persisted on the local filesystem under C:\Skill Bridge\backend\uploads. Ensure this directory is included in system backups to preserve certificate file attachments.

DATA INTEGRITY CRITICAL ADVISORY

Always back up backend\uploads concurrently with MongoDB dumps. If a database record references a certificate filename that does not exist in backend\uploads, the UI will display a missing file alert.

9. Deployment Sign-Off & Maintenance Checklist

Complete this pre-flight checklist prior to handing over the SkillBridge system for college evaluation, viva demonstration, or production deployment:

Verification Item	Verification Standard	Status
MongoDB Service	Service is running and listening on port 27017.	**[VERIFIED]**
FastAPI Microservice	Python venv active; /health returns HTTP 200 on port 8000.	**[VERIFIED]**
Express API Gateway	Node backend running on port 5000; connected to MongoDB and AI service.	**[VERIFIED]**
React Vite Frontend	Vite dev server running on port 5173; clean TypeScript compile.	**[VERIFIED]**
One-Click Desktop Launchers	start_skillbridge.bat and stop_skillbridge.bat tested and working.	**[VERIFIED]**
Demonstration Seed Data	Student, Faculty, Admin, and Recruiter seeded and authenticated.	**[VERIFIED]**
Real Student Verification	omingale26@gmail.com with 12 uploaded certificates validated.	**[VERIFIED]**
RBAC Access Security	403 Forbidden enforced on unauthorized cross-role API calls.	**[VERIFIED]**
Documentation Artifacts	User Manual.pdf and Installation Manual.pdf compiled at project root.	**[VERIFIED]**

Lead Engineer Sign-Off:	SkillBridge Engineering Team
System Architecture Status:	Production Ready / Demo Verified
Evaluation Date:	November 2026